Ex.No: 3

7.08.25 Lab - 3 Study of the classifier with respect to statistical parameter.

Aim:

To implement different classification algorithms for the prediction of breast cancer.

Objective:

1. To load and prepare the Breast cancer dataset.
2. To train and test three ML classifiers:
   1. Decision Tree, SVM, and Logistic Regression
3. To evaluate each model using statistical parameters.

Pseudocode:

1. Decision Tree.
   1. Import breast cancer dataset from Sklearn.
   2. Split the dataset into training and test.
   3. Train the model on the training data.
   4. Predict the test data and find accuracy.

2. **SVM:**
   1. Split the data into training and test.
   2. Train the model on the training data.
   3. Predict the test data label.

Observation:

Classifier	Decision Tree	SVM	Logistic
Accuracy	84.72%	98.61%	97.50%
macro Avg Precision	85.30%	98.72%	97.67%
macro Avg Recall	37.98%	98.66%	97.65%
macro Avg F1-score	84.31%	78.68%	97.65%
weighted Avg F1-score	84.72%	98.61%	97.31%

Observation:

→ Accuracies

1, Decision - 84.72%.

2, SVM - 98.65%.

3, Logistic regression - 97.30%.

Conclusion:

For breast cancer diagnosis where minimizing false negative is critical, SVM is preferred model.

Result:

This shows that SVM and Logistic regression are more effective for high dimensional datasets.

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Code | Notebook | Python 3 (ipykernel)

[1]:
```
!pip install scikit-learn
```

```
Defaulting to user installation because normal site-packages is not writeable
Collecting scikit-learn
  Downloading scikit_learn-1.7.1-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.wh
l.metadata (11 kB)
Requirement already satisfied: numpy>=1.22.0 in ./.local/lib/python3.10/site-packages (fro
m scikit-learn) (2.2.6)
Collecting scipy>=1.8.0 (from scikit-learn)
  Downloading scipy-1.15.3-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.meta
data (61 kB)
     ---------------------------------------- 62.0/62.0 kB 2.3 MB/s eta 0:00:00
Collecting joblib>=1.2.0 (from scikit-learn)
  Downloading joblib-1.5.1-py3-none-any.whl.metadata (5.6 kB)
Collecting threadpoolctl>=3.1.0 (from scikit-learn)
  Downloading threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)
Downloading scikit_learn-1.7.1-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.whl
(9.7 MB)
   ---------------------------------------- 9.7/9.7 MB 1.4 MB/s eta 0:00:0000:0100:01
Downloading joblib-1.5.1-py3-none-any.whl (307 kB)
```

[2]:
```
!pip install --upgrade pip
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pip in /opt/tljh/user/lib/python3.10/site-packages (24.0)
Collecting pip
  Downloading pip-25.2-py3-none-any.whl.metadata (4.7 kB)
Downloading pip-25.2-py3-none-any.whl (1.8 MB)
   ---------------------------------------- 1.8/1.8 MB 7.6 MB/s eta 0:00:00a 0:00:01
Installing collected packages: pip
```

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lab2.ipynb (6) - JupyterLab | Breast Cancer Wisconsin (Diagn | Why use load_digits | Inbox (2,287) - rh0591@srmist

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Code | Notebook | Python 3 (ipykernel)

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Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pip in /opt/tljh/user/lib/python3.10/site-packages (24.0)
Collecting pip
  Downloading pip-25.2-py3-none-any.whl.metadata (4.7 kB)
Downloading pip-25.2-py3-none-any.whl (1.8 MB)
   ---------------------------------------- 1.8/1.8 MB 7.6 MB/s eta 0:00:00a 0:00:01
Installing collected packages: pip
  WARNING: The scripts pip, pip3 and pip3.10 are installed in '/home/jupyter-ra231104701001
0/.local/bin' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --n
o-warn-script-location.
Successfully installed pip-25.2

[notice] A new release of pip is available: 24.0 -> 25.2
[notice] To update, run: pip install --upgrade pip
```

```
[3]: from sklearn.datasets import load_digits
```

```
[4]: d=load_digits()
```

```
[5]: x=d.data
     y=d.target
```

```
[6]: x
```

```
[6]: array([[ 0.,  0.,  5., ...,  0.,  0.,  0.],
            [ 0.,  0.,  0., ..., 10.,  0.,  0.],
            [ 0.,  0.,  0., ..., 16.,  9.,  0.],
            ...,
```

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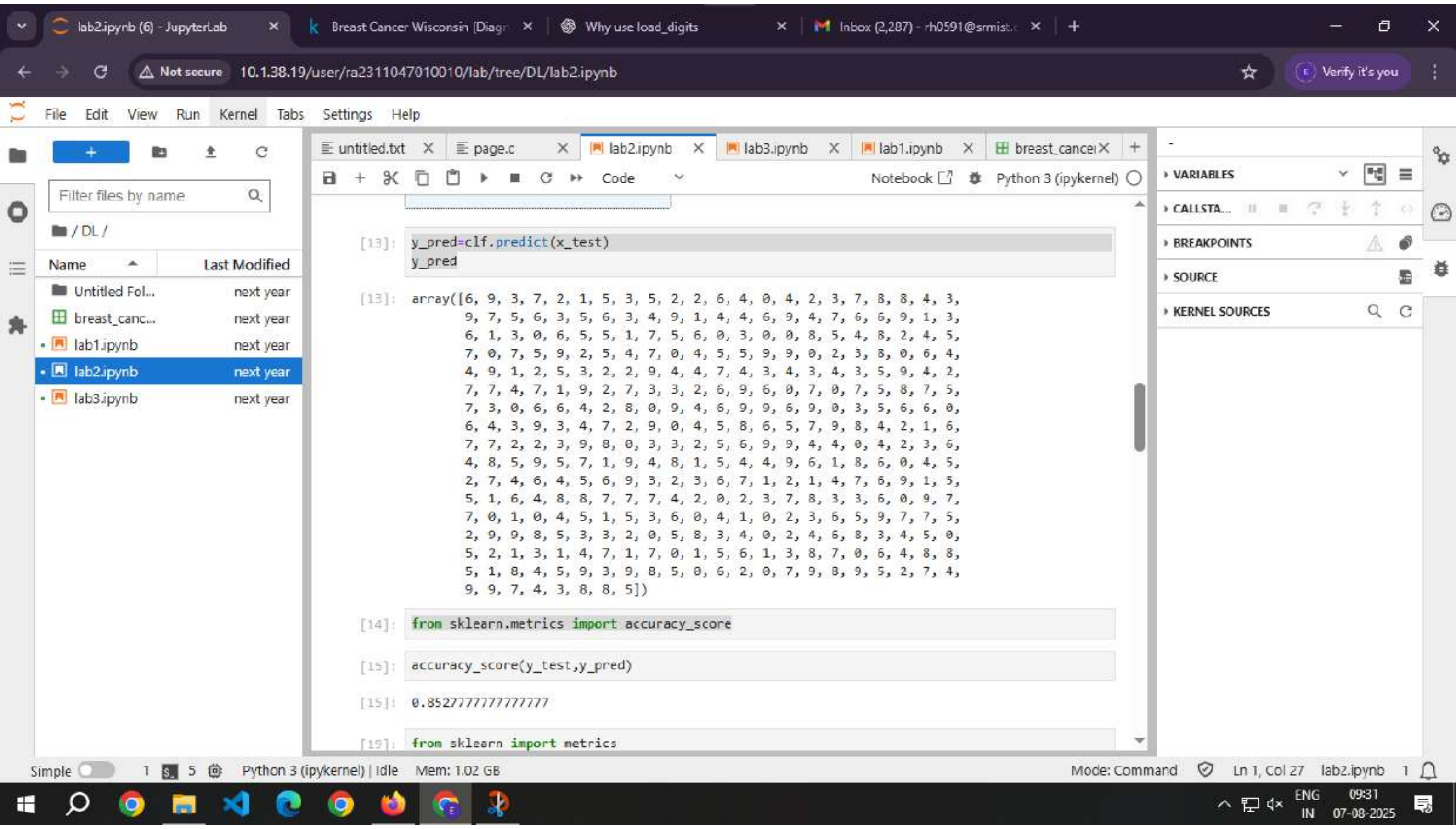
lab2.ipynb (6) - JupyterLab
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breast_cancer
Code
Notebook
Python 3 (ipykernel)
[13]: y_pred=clf.predict(x_test)
y_pred
[13]: array([6, 9, 3, 7, 2, 1, 5, 3, 5, 2, 2, 6, 4, 0, 4, 2, 3, 7, 8, 8, 4, 3,
9, 7, 5, 6, 3, 5, 6, 3, 4, 9, 1, 4, 4, 6, 9, 4, 7, 6, 6, 9, 1, 3,
6, 1, 3, 0, 6, 5, 5, 1, 7, 5, 6, 0, 3, 0, 0, 8, 5, 4, 8, 2, 4, 5,
7, 0, 7, 5, 9, 2, 5, 4, 7, 0, 4, 5, 5, 9, 9, 0, 2, 3, 8, 0, 6, 4,
4, 9, 1, 2, 5, 3, 2, 2, 9, 4, 4, 7, 4, 3, 4, 3, 4, 3, 5, 9, 4, 2,
7, 7, 4, 7, 1, 9, 2, 7, 3, 3, 2, 6, 9, 6, 0, 7, 0, 7, 5, 8, 7, 5,
7, 3, 0, 6, 6, 4, 2, 8, 0, 9, 4, 6, 9, 9, 6, 9, 0, 3, 5, 6, 6, 0,
6, 4, 3, 9, 3, 4, 7, 2, 9, 0, 4, 5, 8, 6, 5, 7, 9, 8, 4, 2, 1, 6,
7, 7, 2, 2, 3, 9, 8, 0, 3, 3, 2, 5, 6, 9, 9, 4, 4, 0, 4, 2, 3, 6,
4, 8, 5, 9, 5, 7, 1, 9, 4, 8, 1, 5, 4, 4, 9, 6, 1, 8, 6, 0, 4, 5,
2, 7, 4, 6, 4, 5, 6, 9, 3, 2, 3, 6, 7, 1, 2, 1, 4, 7, 6, 9, 1, 5,
5, 1, 6, 4, 8, 8, 7, 7, 7, 4, 2, 0, 2, 3, 7, 8, 3, 3, 6, 0, 9, 7,
7, 0, 1, 0, 4, 5, 1, 5, 3, 6, 0, 4, 1, 0, 2, 3, 6, 5, 9, 7, 7, 5,
2, 9, 9, 8, 5, 3, 3, 2, 0, 5, 8, 3, 4, 0, 2, 4, 6, 8, 3, 4, 5, 0,
5, 2, 1, 3, 1, 4, 7, 1, 7, 0, 1, 5, 6, 1, 3, 8, 7, 0, 6, 4, 8, 8,
5, 1, 8, 4, 5, 9, 3, 9, 8, 5, 0, 6, 2, 0, 7, 9, 8, 9, 5, 2, 7, 4,
9, 9, 7, 4, 3, 8, 8, 5])
[14]: from sklearn.metrics import accuracy_score
[15]: accuracy_score(y_test,y_pred)
[15]: 0.8527777777777777
[19]: from sklearn import metrics
VARIABLES
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SOURCE
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```
[22]: import matplotlib.pyplot as plt
```

```
[23]: plt.show()
```

```
[24]: plt.matshow(d.images[5], cmap="gray")
```

```
[24]: <matplotlib.image.AxesImage at 0x71d129ea11b0>
```